

Compatibility of spectral and operator-preserving tight-binding reductions of a first-principles silicon Hamiltonian

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Abstract

First-principles electronic Hamiltonians may be reduced to compact lattice models by preserving either selected spectral properties or an aligned localized operator. These criteria need not admit the same reduced representation. This study formulates their compatibility within a common hierarchy of orthogonal sp^3s^* Slater–Koster models for bulk silicon. A validated ten-orbital Wannier Hamiltonian will be constructed from one converged density-functional-theory calculation. For each tight-binding model class, the spectral and operator criteria define admissible sets satisfying declared error tolerances. Compatibility requires a nonempty intersection, corresponding to a single Hamiltonian satisfying both requirements. When the intersection is empty, a normalized real-space Hamiltonian metric will determine the minimum separation between the admissible sets and quantify the incompatibility of the prescribed model class with the retained first-principles information. Validation will use withheld band energies, the indirect gap, conduction-valley position, longitudinal and transverse electron effective masses, and operator residuals resolved by orbital block and neighbor shell. The study will identify the smallest tested model class admitting a common reduced representation satisfying both criteria and thereby establish a validated bulk reference for subsequent impurity-operator reduction.

Keywords: silicon, Wannier Hamiltonian, tight-binding reduction, operator

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1. Introduction

Density-functional theory (DFT) supplies a first-principles Kohn–Sham description of crystalline electronic structure, but its plane-wave representation is inconvenient for interpolation and reduced lattice calculations. Maximally localized Wannier functions provide a localized representation of a selected Kohn–Sham subspace and support accurate interpolation [1, 2, 3, 4]. Parameterized tight-binding models instead restrict the Hamiltonian to a prescribed orbital basis, hopping range, and symmetry structure [5, 6].

A spectral fit preserves selected eigenvalues, whereas an operator fit preserves matrix information in an aligned localized representation. Spectral agreement alone does not establish agreement of onsite terms, orbital couplings, or hopping amplitudes. These differences become consequential when the bulk Hamiltonian is later modified by dopants, defects, strain, or other local perturbations, since all such operators must ultimately be represented in the same reduced basis to form a consistent effective Hamiltonian. Before an operator residual can be evaluated, the Wannier and tight-binding orbital coordinates must therefore be identified through a stable, symmetry-compatible alignment [7, 8].

The present study asks whether the spectral and operator criteria admit a common reduced representation within a prescribed tight-binding model class. Rather than forcing two independently fitted parameter vectors to agree, it defines admissible model sets for each criterion and asks whether they intersect. Repeating this test over a nested hierarchy of tight-binding model classes identifies the smallest class admitting a common representation, or quantifies the obstruction when no compatible representation exists.

2. Methodology

2.1. First-principles reference calculation

The first-principles reference will be generated using Quantum ESPRESSO [9]. Exchange and correlation will be treated using the Perdew–Burke–Ernzerhof (PBE) functional [10]. An optimized norm-conserving pseudopotential from PseudoDojo [11, 12] will be used in scalar-relativistic form. The parent problem will use the PBE-relaxed lattice constant and

will be non-spin-polarized without spin-orbit coupling. Plane-wave cutoffs, wavevector meshes, and the number of bands will be converged against the indirect gap, conduction-valley position, and longitudinal and transverse electron effective masses.

2.2. Self-consistent ground-state calculation

The physical purpose of the initial self-consistent-field (SCF) calculation is to determine the ground-state electron density and hence the effective one-electron Kohn-Sham operator [13]. Mathematically, this is a nonlinear fixed-point problem because the operator depends on the density constructed from its eigenstates. At iteration i , the cycle is

$$n^{(i)}(\mathbf{r}) \longrightarrow V_{\text{KS}}[n^{(i)}](\mathbf{r}) \longrightarrow \hat{H}_{\text{KS}}[n^{(i)}] \longrightarrow \left\{ \epsilon_{n\mathbf{k}}^{(i)}, \psi_{n\mathbf{k}}^{(i)} \right\} \longrightarrow n^{(i+1)}(\mathbf{r}), \quad (1)$$

where $n^{(i)}(\mathbf{r})$ is the density, $V_{\text{KS}}[n^{(i)}](\mathbf{r})$ is the corresponding effective Kohn-Sham potential, $\hat{H}_{\text{KS}}[n^{(i)}]$ is the density-dependent Kohn-Sham operator, and $\psi_{n\mathbf{k}}^{(i)}$ and $\epsilon_{n\mathbf{k}}^{(i)}$ are the orbital and eigenvalue for band n and wavevector $\mathbf{k}$. Here $n^{(i+1)}(\mathbf{r})$ denotes the next input density after reconstructing an output density from the occupied orbitals and applying the selected mixing; it is not generally the unmixed reconstructed density. Numerically, the cycle uses iterative diagonalization, occupations, density reconstruction, and mixing until the code-specific SCF convergence criterion is satisfied. The converged quantities are denoted

$$n_{\text{SCF}}(\mathbf{r}) \quad \text{and} \quad V_{\text{KS}}[n_{\text{SCF}}](\mathbf{r}). \quad (2)$$

No particular density norm is assumed here.

The principal downstream role of the SCF calculation is therefore to fix the effective operator. A non-self-consistent-field (NSCF) or band calculation holds the converged density and potential fixed while solving

$$\hat{H}_{\text{KS}}[n_{\text{SCF}}]\psi_{n\mathbf{k}} = \epsilon_{n\mathbf{k}}\psi_{n\mathbf{k}} \quad (3)$$

on a wavevector set chosen for the observable. SCF sampling is selected to converge the density and total energy; an NSCF mesh may instead support Brillouin-zone integration or state extraction, whereas a symmetry path is selected for dispersion visualization and symmetry-resolved analysis. Effective-mass, valley, Wannier, and tight-binding calculations can consequently require different meshes, retained bands, and convergence studies. These calculations are not interchangeable merely because they use the same converged potential.

Before production dataset design, a single-process Quantum ESPRESSO 7.2 `pw.x` silicon tutorial SCF calculation was used as a bounded computational bootstrap [9]. Its identified legacy tutorial pseudopotential is distinct from the planned PBE/PseudoDojo production parent described above. The bootstrap established the executable and pseudopotential identities, successful invocation, input/output provenance, reproduction of the distributed tutorial observation, and creation and inspection of restart and QEXSD artifacts. It also established which metadata were actually available from the backend and supported the separation of QEXSD input/output mechanics, periodic geometry, Kohn–Sham semantics, and plane-wave record ownership, with explicit units, coordinate conventions, reciprocal scaling, and unavailable metadata. The retained observation was

$$E_{\text{SCF}} = -15.84452726 \text{ Ry}, \quad (4)$$

with convergence reported after six iterations and agreement with the bundled reference at printed precision. This is execution and extraction evidence, not numerical or physical validation. The limited extraction claim was human-accepted, while the resulting record remains software-verification evidence rather than a human-accepted scientific dataset. The corresponding compact provenance and record architecture are retained with the repository calculation record and computational documentation.¹

The tutorial calculation is not the production silicon dataset. Its ten sampled wavevectors and four retained bands were sufficient for execution and extraction verification, but not for production convergence, determination of the indirect gap, conduction-valley location or curvature, effective-mass extraction, Wannierization, tight-binding fitting, physical validation, or uncertainty quantification. The workflow boundary is therefore

tutorial SCF $\longrightarrow$ observed QE artifacts
 $\longrightarrow$ artifact inventory $\longrightarrow$ QEXSD semantic extraction
 $\longrightarrow$ human-accepted extraction
 $\longrightarrow$ retained plane-wave KS record
 $\longrightarrow$ separately designed NSCF/band calculations.

¹Repository paths: `calculations/bulk-silicon/qe-example01-si-scf-davidson/` and `docs/computational/ksdft-pw-record-architecture.md`.

The bootstrap verifies implementation and provenance boundaries; the later SCF, NSCF, band, Wannier, and fitting calculations must be designed and converged for their separate scientific purposes.

2.3. Wannier representation

A ten-orbital target subspace containing the valence bands and the low conduction states required for the silicon valleys will then be constructed with Wannier90 [2, 3]. Its real-space Hamiltonian is

$$[\mathbf{H}_W(\mathbf{R})]_{\alpha\beta} = \langle w_{\alpha\mathbf{0}} | \hat{H}_{\text{KS}} | w_{\beta\mathbf{R}} \rangle, \quad (5)$$

where $\mathbf{R}$ is a direct-lattice translation; α and β label Wannier orbitals in the reference and translated primitive cells, respectively; $|w_{\alpha\mathbf{0}}\rangle$ and $|w_{\beta\mathbf{R}}\rangle$ are localized Wannier states; and $\hat{H}_{\text{KS}}$ is the converged Kohn–Sham operator.

Initial projections and outer and frozen disentanglement windows will be recorded. The Wannier Hamiltonian will be accepted as the common parent only after its dense-grid interpolation error, centers, spreads, sensitivity to window and projection choices, and real-space hopping decay have been validated.

2.4. Spectral and operator losses

For each candidate model class, the spectral loss is a weighted error over training-set Wannier eigenvalues and selected band-edge quantities. The operator loss is

$$\mathcal{L}_H(\mathbf{C}, \boldsymbol{\theta}) = \sum_{\mathbf{R}} \omega_{\mathbf{R}} \|\mathbf{H}_W(\mathbf{R}) - \mathbf{C}\mathbf{H}_{\text{TB}}(\mathbf{R}; \boldsymbol{\theta})\mathbf{C}^\dagger\|_{\text{F}}^2, \quad (6)$$

where $\mathbf{C} \in \mathcal{U}$ is a symmetry-compatible unitary alignment from the tight-binding orbital coordinates to the Wannier coordinates; $\boldsymbol{\theta}$ is the tight-binding parameter vector; $\mathbf{H}_{\text{TB}}(\mathbf{R}; \boldsymbol{\theta})$ is the tight-binding hopping matrix associated with $\mathbf{R}$; $\omega_{\mathbf{R}} \geq 0$ weights the contribution of each retained neighbor shell; and $\|\cdot\|_{\text{F}}$ is the Frobenius norm.

2.5. Tight-binding model hierarchy

The model hierarchy will begin with an orthogonal ten-orbital sp^3s^* Hamiltonian with nearest-neighbor hopping. If that class is incompatible with the declared tolerances, second-neighbor and selected symmetry-allowed corrections will be introduced in a fixed sequence.

For a candidate model class $\mathfrak{M}_j$, the two losses define the admissible sets

$$\mathfrak{A}_{E,j}(\tau_E) = \{ \mathbf{H}(\boldsymbol{\theta}) \in \mathfrak{M}_j : \mathcal{L}_E(\boldsymbol{\theta}) \leq \tau_E \}, \quad (7)$$

$$\mathfrak{A}_{H,j}(\tau_H) = \left\{ \mathbf{H}(\boldsymbol{\theta}) \in \mathfrak{M}_j : \min_{\mathbf{C} \in \mathcal{U}} \mathcal{L}_H(\mathbf{C}, \boldsymbol{\theta}) \leq \tau_H \right\}, \quad (8)$$

where $\mathcal{L}_E$ is the spectral loss; $\mathcal{L}_H$ is the aligned operator loss; τ_E is the declared spectral tolerance; τ_H is the declared operator tolerance; and j indexes the nested tight-binding model classes.

The spectral tolerance includes targets of 5% for the indirect gap, 3% of the Γ - X distance for the conduction-valley position, and 8% for each electron effective mass. Operator and alignment tolerances will be normalized to the converged Wannier Hamiltonian and fixed before the compatibility search.

2.6. Compatibility metric

A dimensionless real-space Hamiltonian metric d_{TB} will compare models in their common canonical tight-binding coordinates. The minimum distance between the two admissible sets is

$$\delta_j^* = \inf_{\substack{\mathbf{H}_E \in \mathfrak{A}_{E,j} \\ \mathbf{H}_H \in \mathfrak{A}_{H,j}}} d_{\text{TB}}(\mathbf{H}_E, \mathbf{H}_H), \quad (9)$$

where $\mathbf{H}_E$ and $\mathbf{H}_H$ are models satisfying the spectral and operator criteria, respectively. A value of δ_j^* below the declared map tolerance establishes compatibility within $\mathfrak{M}_j$; otherwise, it quantifies the irreducible separation under the imposed spectral and operator requirements.

All accepted models will be tested on withheld wavevectors. The operator residual will be decomposed by onsite and hopping contribution, orbital block, crystal-symmetry channel, and neighbor shell. The analysis is restricted to bulk silicon; impurity operators are outside the present study.

All spectral and operator errors are measured relative to the converged PBE/Wannier parent Hamiltonian. The present study tests reduction fidelity and does not treat agreement with the experimental silicon band gap as an independent fitting target.

3. Expected results and decision criteria

The required outputs are one converged bulk-silicon DFT dataset, one validated ten-orbital Wannier Hamiltonian, admissible spectral and operator

Table 1: Decision criteria for the compatibility analysis.

Quantity	Mathematical criterion	Decision role
Spectral admissibility	Spectral loss within threshold	Retains selected bands and band-edge quantities
Operator admissibility	Aligned operator loss within threshold	Retains real-space matrix information
Map compatibility	Set distance within map threshold	Establishes a common reduced Hamiltonian
Model selection	Lowest-order compatible class	Selects the minimal common Hamiltonian class

model sets for each tested tight-binding class, their minimum metric separation, and one common validation record. Numerical entries will be reported only after the convergence, fitting, and sensitivity calculations are complete.

The decisive result is the smallest tested model class for which the spectral and operator admissible sets intersect within tolerance. If no tested class is compatible, the nonzero set separation and its decomposition will identify the orbital or hopping content responsible for the obstruction. Either outcome provides a controlled basis for determining whether the chosen reduced representation can consistently support subsequent dopant, defect, or impurity operators.

This work addresses compatibility for a single pair of reduction criteria. The same framework naturally extends to additional operator families, where the objective is to determine whether a common reduced representation exists for the bulk Hamiltonian together with perturbation operators describing dopants, defects, strain, or electron-phonon interactions.

CRedit authorship contribution statement

Bryan D. Llenarizas: Formal analysis, Investigation, Validation, Data curation, Visualization, Writing—original draft, Writing—review and editing. Eugene Joseph M. Ragasa: Conceptualization, Methodology, Software, Validation, Supervision, Project administration, Writing—review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The versioned source code, computational inputs, processed datasets, and reproduction workflows supporting the final article will be deposited in persistent repositories. Their permanent identifiers will be inserted here before submission.

Declaration of generative AI and AI-assisted technologies

During preparation of this work, the authors used ChatGPT and Codex for drafting assistance, software-development support, and editorial review. The authors reviewed and edited all resulting content and take full responsibility for the content of the publication.

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